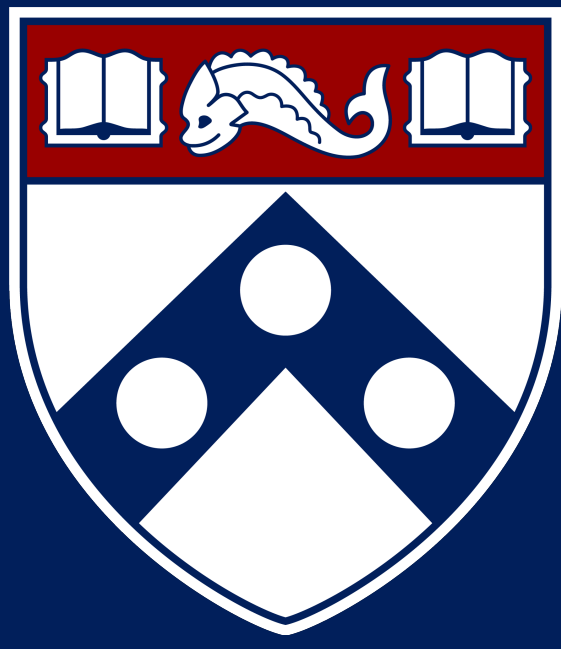


Why do marked words persist?



Corpus and experimental evidence for competition between phonological and semantic well-formedness in word-formation and the lexicon

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Takeaway

A phonologically marked form survives in a language's lexicon & word formation when there are fewer semantically equivalent alternative forms.

Background

In probabilistic grammar, marked forms tend to become **marginalized** and die out over time. However, there are exceptions where marked forms are **tolerated** [1, 2, 3].

The mystery of survival:

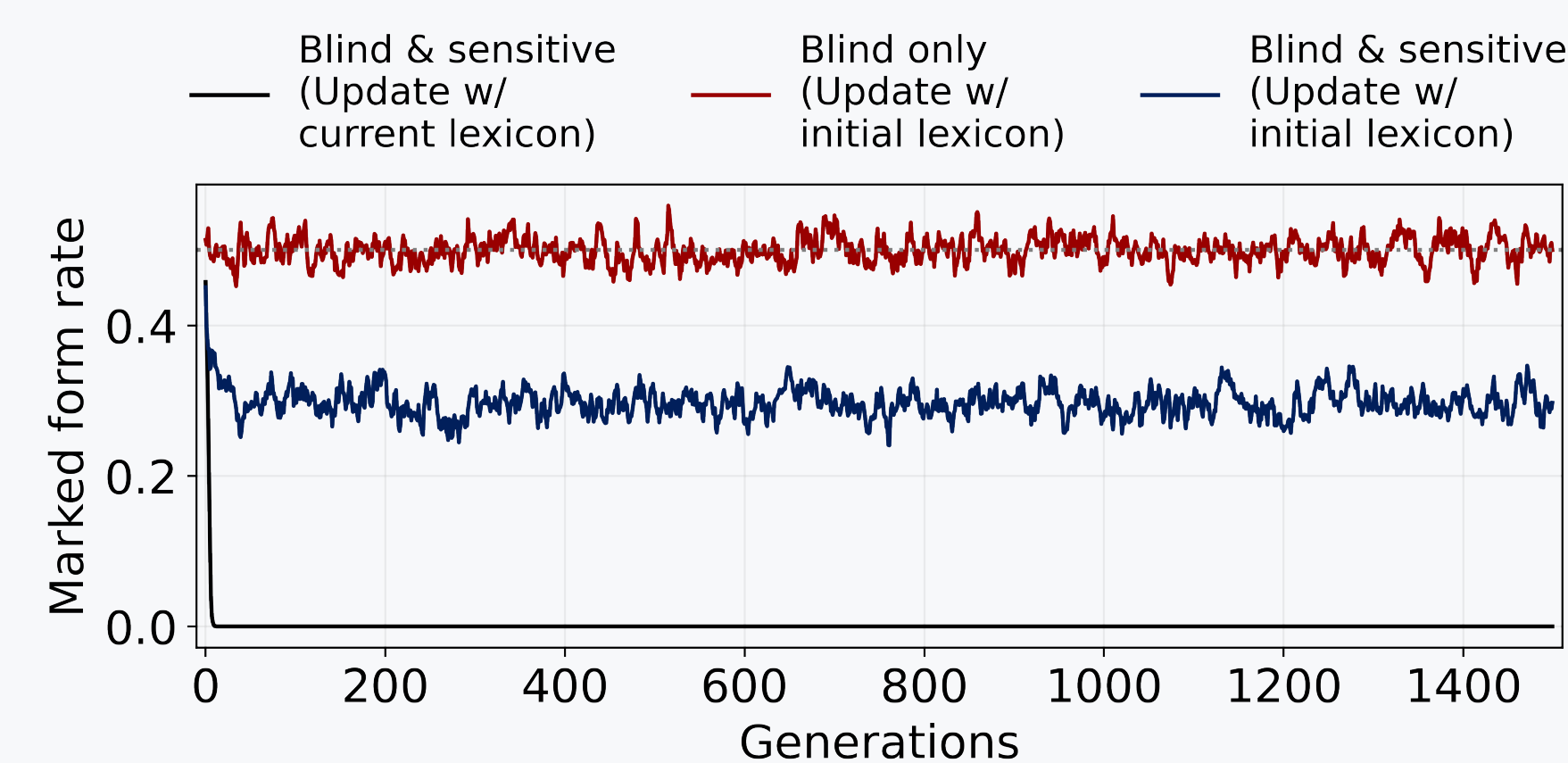
- If marked forms are dispreferred, why do they survive in the lexicon?
- Martin's simulation [2]: when a grammar is updated from the attested distribution, marked forms **die out** over generations.

Methods

MaxEnt models learned from the previous generation & produced new data for the following generation.

Table 1. Training data at initialization.

	CC	C+C
pt	100%	50%
pp	0%	50%



Hypothesis

Word form flexibility (FLEX) competes with phonological well-formedness

FLEX= N of alternative forms compatible with the target semantics.

Corpus studies

Language	Constraint	Example
English	*GEMINATE	<i>bookkeeper</i>
Finnish	*[a _{back}][_{-a_{back}}]	<i>kesä_{back}lomaa</i> "summer vacation"
Hungarian	*[a _{back}][_{-a_{back}}]	<i>kő_{back}por</i> "rock flour"

Data two-member compounds from Wiktionary and published datasets [4,5]

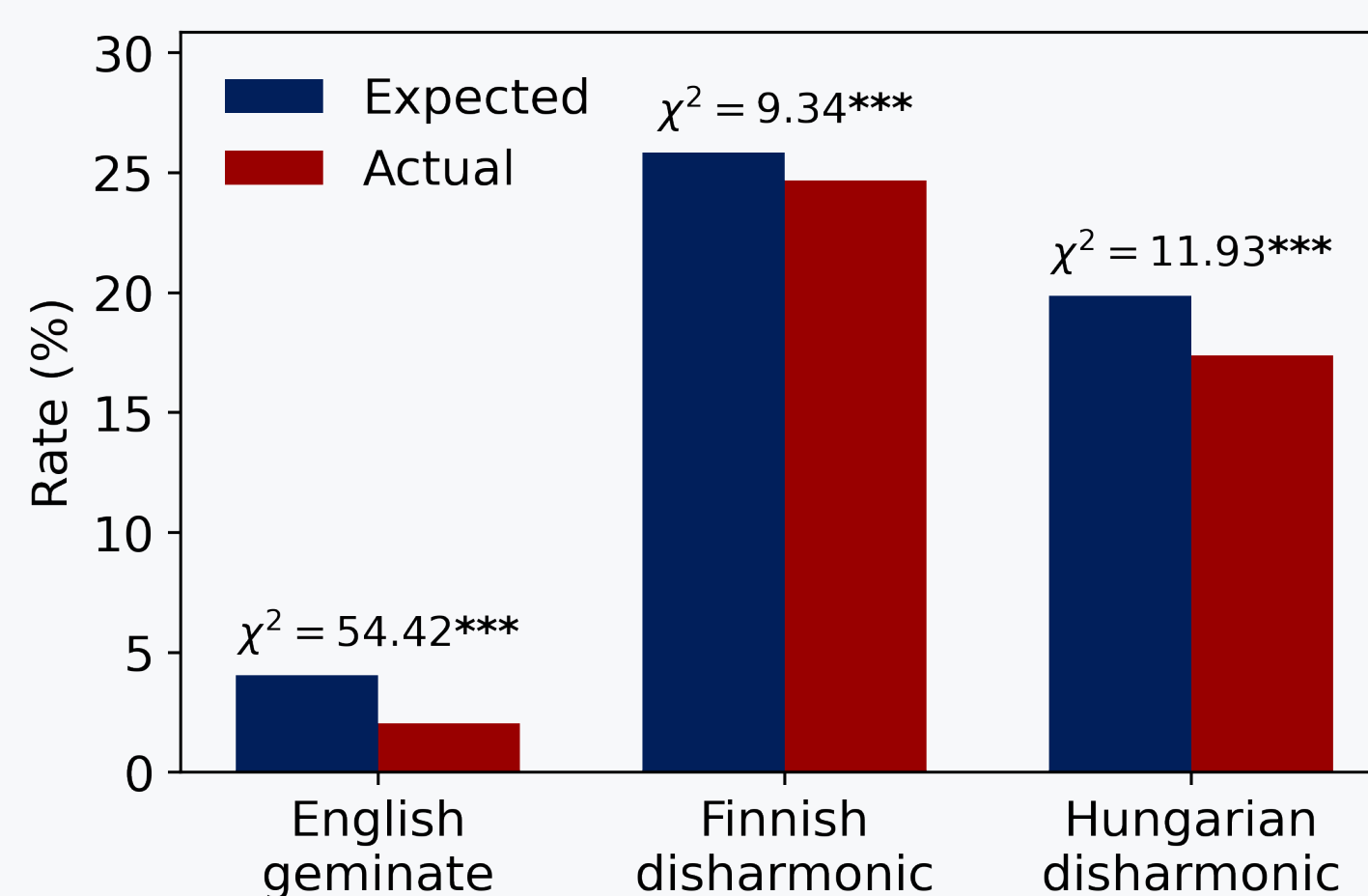
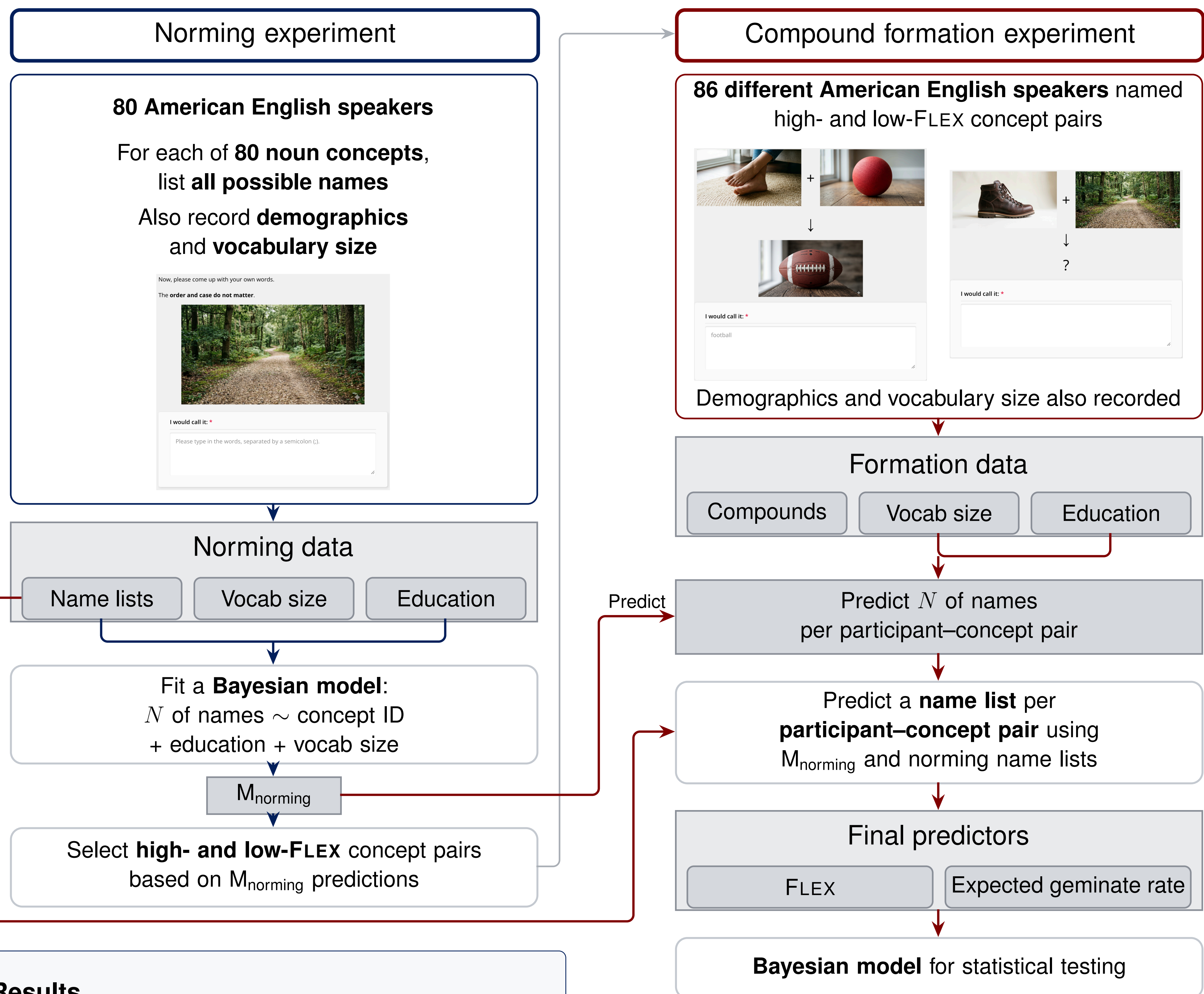


Figure 1. Geminate & disharmonic compounds are under-represented.

Methods

- $FLEX(compound) = FLEX(member1) \times FLEX(member2)$
- $FLEX(word) = \text{cosine similarity of the word's vector w/ all word vectors in the language}$



Results

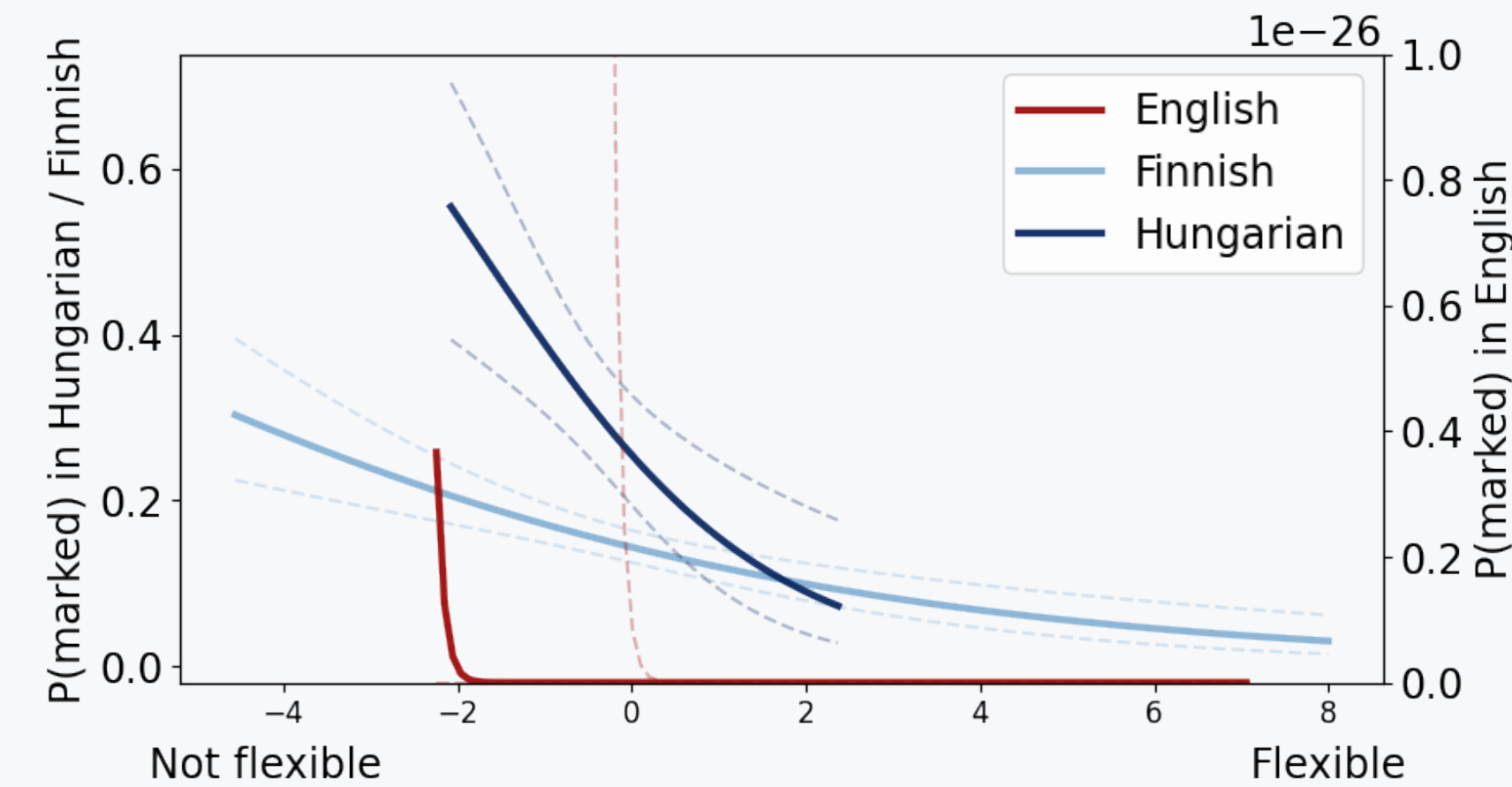


Figure 2. FLEX effects on a compound's markedness.

Examining diachronic evolution through MaxEnt simulation

FLEX:

For each word:

- $k-1$ candidate forms were created, w/ the same geminate rates.
- k was randomly sampled from a Zipfian distribution (exponent= s).

Results

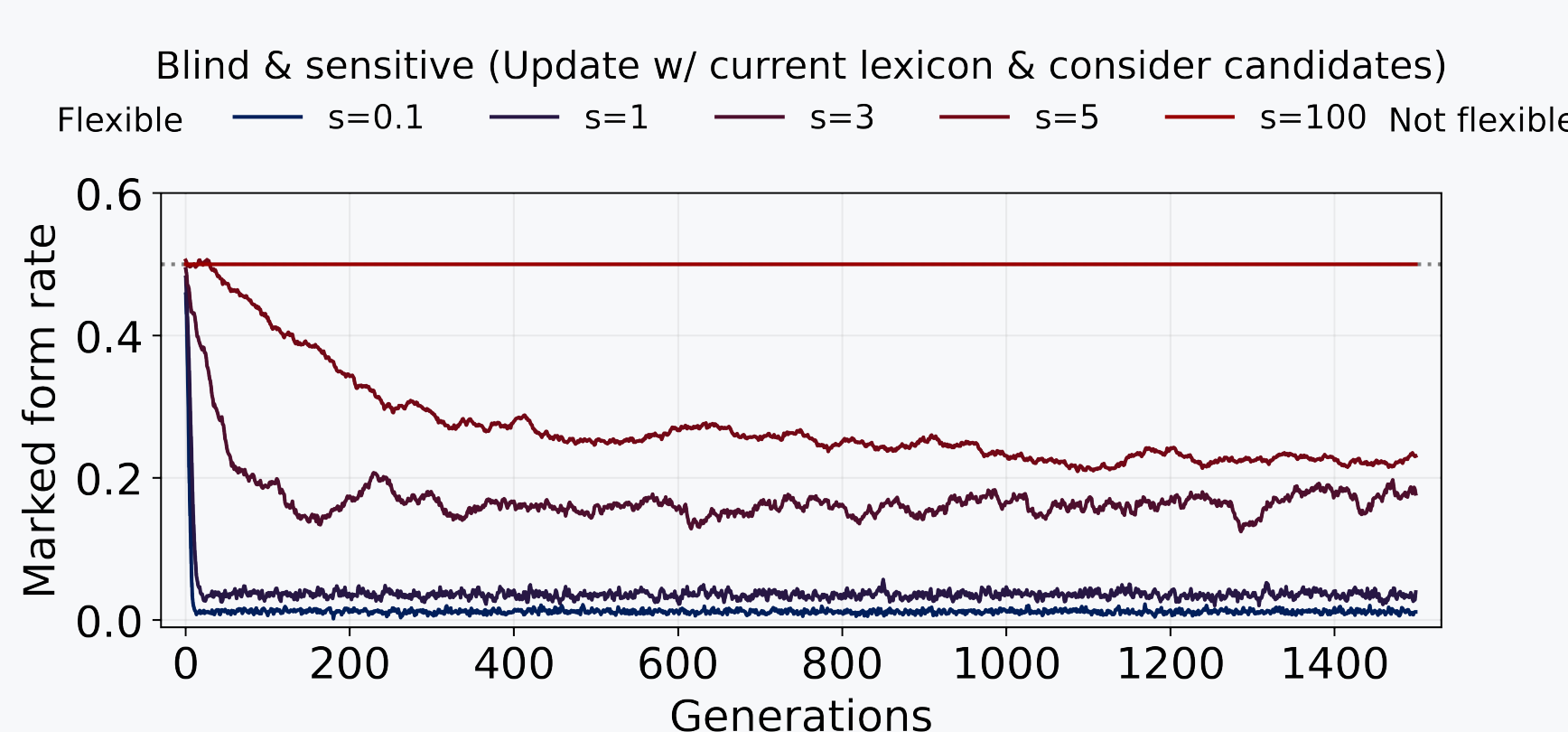


Figure 3. Simulated evolution of fake geminate compounds in the lexicon.

Compound formation experiment

Results

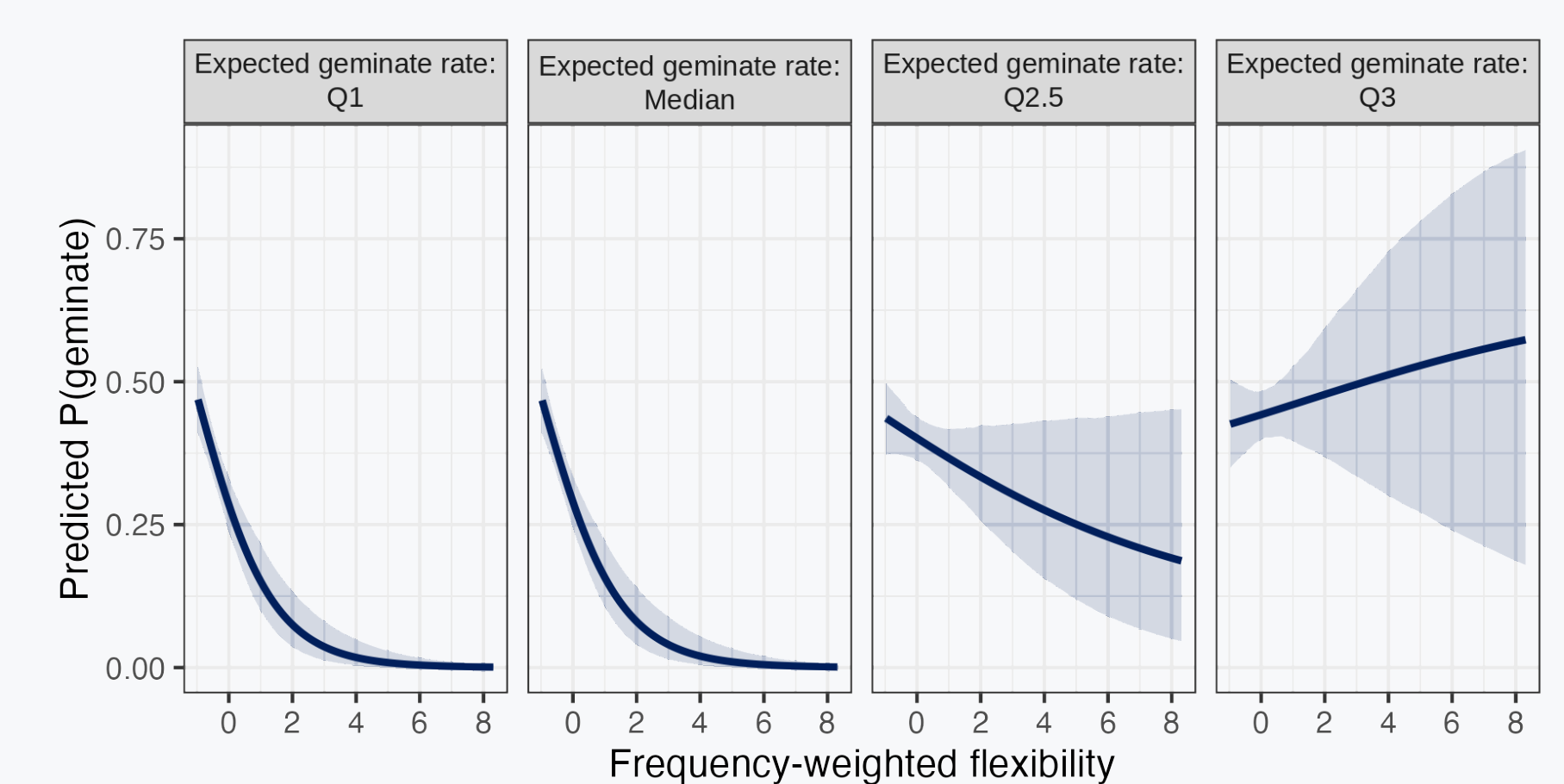


Figure 5. Model-derived interaction b/t a compound's FLEX & the markedness rate of the name offered.

Selected references

- [1] Kempen, G., & Harbusch, K. (2008). Comparing linguistic judgments and corpus frequencies as windows on grammatical competence. In A. Steube (Ed.), *The discourse potential of underspecified structures* (pp. 179–192). De Gruyter.
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Acknowledgment

We thank Yue Chen and Linh Pham for providing directions during the design of the compound formation experiment and the PhonLunch@USC audience for their feedback on an earlier version of this study.